SEEING CHANGE FROM ORBIT · MEETING MINUTES

Meeting 4: Registration, Segmentation And Task Allocation

Date and time	Mon 28 Sep 2026, 1:30–2:30 pm
Type	Supervisor
Location	Online (MS Teams)
Attendees	Dr. Panos, Zain, Ayesha, Khaled, Tehsin, Zayed

Agenda

1. Fixes since last meeting: 6-band Sentinel-2 and data dictionary (Zain)
2. Next step: registration or segmentation?
3. Spectral bands and mangrove constraints
4. Reference papers
5. Gantt chart
6. Task allocation and GitHub

Member updates

Who	Update
Zain	6-band Sentinel-2 imagery for Jubail (2018, 2025), first QGIS work, data dictionary
Khaled	Draft Gantt chart
Tehsin	Found the DECDNet reference paper, adopted as the Track B blueprint
Ayesha	Segmentation literature review; raised that registration needed study before segmentation

Discussion

1. Six-band imagery and data dictionary

- Zain showed 6-band Sentinel-2 (RGB, NIR, SWIR) for Jubail, 2018 and 2025, his first QGIS work, and a data dictionary with source and file metadata.

2. Registration before segmentation

- Zain asked whether to build a model and compare with GMW next. Dr. Panos: register the images first.
- Method: grayscale from RGB → pick a reference image → detect corners automatically → match points → fit an affine transform by optimisation → deform the other image. Test with multiple landmarks and on multiple image pairs.
- Landmark points can also stitch adjacent windows into mosaics and align mosaics across years.
- Registration and segmentation can run in parallel. Limit registration to two weeks.

3. Segmentation, bands and constraints

- After registration: use the six bands to find coastlines, segment mangroves, label each region, and count pixels/area for both years.

- Ayesha: the literature review covered detection methods, not which bands describe vegetation. Dr. Panos wants several published reference tables on vegetation bands, compared with the Sentinel-2 metadata.
- Use prior knowledge: keep green areas within a distance of the coast and next to sea (from blue/IR).
- Clustering the six bands is only naive unsupervised segmentation. Use supervised learning with ground truth, geographically broad imagery, and train/validation/test splits.
- Split segmentation research: conventional ML, U-Net, foundation models (e.g. SAM). Adapt pretrained models; don't train from scratch.

4. Reference papers

- Dr. Panos showed his 2023 marine-debris segmentation work (class imbalance, imperfect predictions): use published weights, hand-label mangrove data, fine-tune, test on held-out data, measure performance.
- Reviewed a 2024 UAE mangrove-monitoring paper: appears to use GMW, Google Earth and visual interpretation; little technical detail; peer review unclear. Khaled to check its labelling and validation.

5. Gantt chart

- Khaled's Gantt chart reviewed: task descriptions need to be clearer and more technical; some durations need adjusting.

6. Task allocation and GitHub

- Dr. Panos wants a responsibility list covering data, segmentation, correspondence, landmark detection and alignment.
- Send the agenda in advance (the morning of the meeting at the latest) so he can prepare.
- GitHub: restricted area for working material; open area for deliverables, code, data and individual contributions. Don't upload papers you don't hold copyright for.
- Ask questions in the Teams channel or by email between meetings; follow up by email if no reply within a day.

Decisions

- Register images before segmenting; validate registration on multiple pairs; max two weeks.
- Registration method: corner detection → matched points → affine transform, on grayscale images with a reference date.
- Segmentation is supervised and fine-tunes pretrained models; three routes (conventional ML, U-Net, foundation model).
- Only green areas near the coast and adjacent to sea count as mangrove candidates.
- Agenda sent before every meeting; questions via Teams/email between meetings.
- GitHub split into restricted and open areas; no copyrighted papers.
- Allocation written up afterwards: Track A = Khaled (landmarks) and Tehsin (alignment, 2 vs 3 landmark error); Track B = Ayesha (SAM, DECDNet), Zayed (U-Net), Zain (RF/SVM); data = Zain (exports, data dictionary), Zayed and Ayesha (GMW ground truth). Weekly meetings Mondays 1:30 on Teams.

Action items

Who	What	Due
Khaled	Landmark detection: corners, filtering noisy corners	~2 weeks
Tehsin	Affine alignment; registration error with 2 vs 3 landmarks	~2 weeks
Ayesha	SAM route; lead DECDNet reading	Not set
Zayed	U-Net route	Not set
Zain	RF/SVM baseline; 6-band exports for more sites; data dictionary	Not set
Zayed, Ayesha	Find GMW ground truth or plan manual labelling	Not set
Khaled	Check the 2024 UAE paper's labelling and validation	Not set
Team	Literature tables on vegetation bands vs Sentinel-2 metadata	Not set
Team	Revise the Gantt chart with clearer technical tasks	Not set
Team	Responsibility list; agenda before each meeting	Next meeting

Next meeting

Mon 5 Oct 2026, 1:30 pm (Teams).